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Stress Testing and Patch Review:

Learning Things The Hard Way

Tales of Three Newbies

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- Newbie #1:
 - Designed/implemented important Linux-kernel functionality ***while a student***, requires only vague guidance, often better results than me
- Newbie #2:
 - Didn't know much, but he knew that he knew RCU better than I did. However, learned very quickly and eventually made great contributions
- Newbie #3:
 - Required constant guidance and direction, often forgetting earlier guidance and often sending nonsensical patches. Made a few small contributions

Tales of Three Newbies

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- Newbie #2: Good return on time invested
- Newbie #3: Negative return on time invested

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You want to be like Newbie #1 or #2
And so does your LLM!!!

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- AI notices “dev” and decides that what I **really** want is to look up the entertainer named Dev

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- ... for **seriously** annoying a user

LLM Version of Newbie #3

- I am trying to copy and paste a command and line
- AI notices “dev” and decides that what I **really** want is to look up the error handler named Dev
- Some AI developers gets a brownie point ...
- ... for seriously annoying a user

Don't let your LLM be like Newbie #3!!!

Fortunately, There *is* Decent AI

- <https://github.com/masoncl/review-prompts>
- <https://sashiko.dev/>
- And many others

And I Have Had Good AI Experiences

- Analyzing code:
 - Can `x()` call `y()`?
 - Does `z()` really need to acquire that lock?
- Repeating work: Like commit A, but on file `f.c`
- Finding obscure Linux-kernel functions:
 - For but one example, `cpumask_next_wrap()`

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 - Can `x()` call `y()`?
 - Does `z()` really need to acquire that lock?
- Repeating work: Like commit A, but on file `f.c`
- Finding obscure linux-kernel functions:
 - For but, for example, `cpumask_next_wrap()`

But even the best AI can hallucinate

On Hallucinations

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 - Who sent in a patch that was mostly nonsense

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- On to (human) Newbie #4...
 - Who sent in a patch that was mostly nonsense
 - Except for one very clever trick that is still in use
 - In this case, I immediately recognized the nonsense
 - LLMs aren't the only things that hallucinate!!!

On Hallucinations

- On to Newbie #5...

On Hallucinations

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- On to Newbie #5... Which would be me.
 - I am not good at recognizing my own nonsense
 - This is why I wrote rcutorture and use many tools
 - “You cannot test correctness into a program”
 - You also cannot verify correctness into a program
 - Your specification might contain bugs

On Hallucinations

- On to Newbie #5... Which would you choose?
 - I am not good at recognizing my own nonsense
 - This is why I wrote recheck and use many tools
 - “You cannot test correctness into a program”
 - You also cannot verify correctness into a program
 - Verification might contain bugs

But testing and verification can (usually) detect nonsense!!!

On Hallucinations

- On to problem #5... Which would you choose?
 - I am not good at recognizing my own nonsense
 - This is why I wrote this presentation can (usually) use many tools
 - “You cannot test very much into a program”
 - You also cannot detect very much into a program
 - Very specific one might contain bugs

Run rcutorture first. Ask questions later.
But testing and verification can (usually) detect no nonsense!!!

On Hallucinations

- On to problem #5... Which would be me.
 - I am not good at recognizing my own nonsense
 - This is why I wrote this presentation. Use many tools
 - “You cannot test a program into a program”
 - You also cannot correct a program into a program
 - Your specification might contain bugs

Run rcutorture first. Hallucination can (usually) be detected by testing and asking questions later.

It is not just me—CI/CD!!!

A Nonsense-Detection Adventure

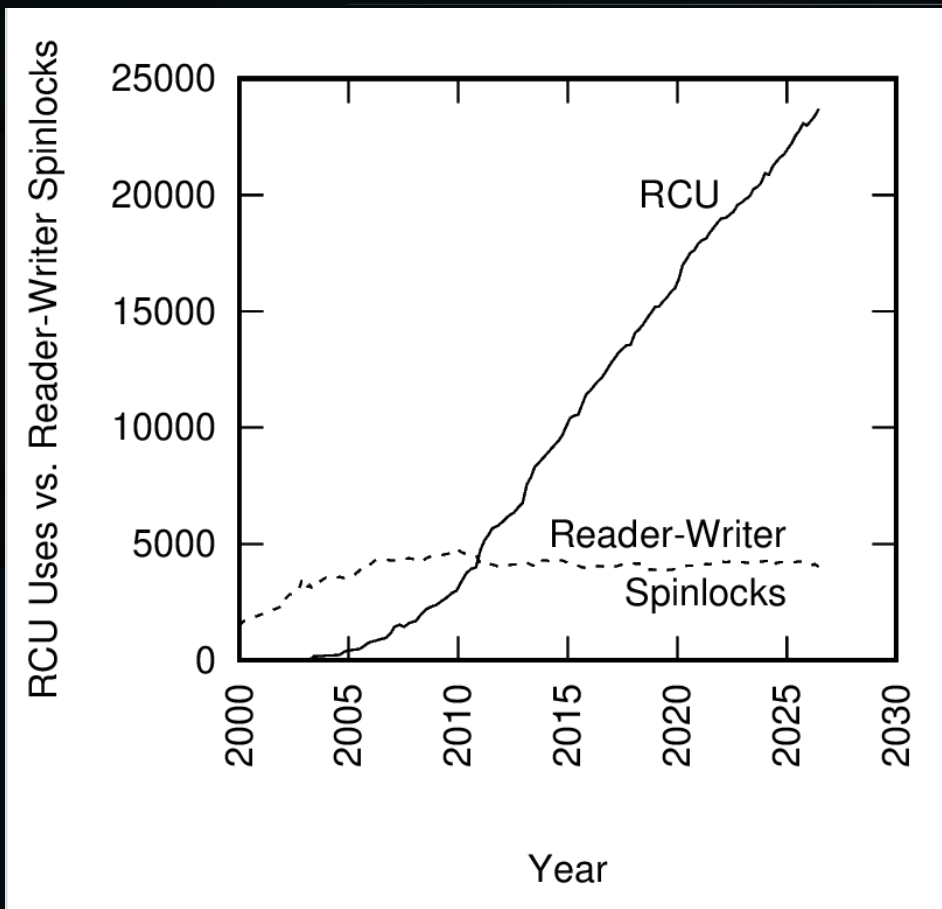
A Nonsense-Detection Adventure



A Nonsense-Detection Adventure

- Shortly after rcutorture was accepted, I got a strange bug report from across the narrow pond
 - Too-short grace periods on a heavily loaded system
 - Every week or so. Sometimes.
 - I asked for the .config file, and got silence
 - Hardware problem? Who knows?

But First, What is RCU???



What is a Too-Short Grace Period???



A Nonsense-Detection Adventure



A Nonsense-Detection Adventure

- Shortly after rcutorture was accepted, I got a strange bug report from across the narrow pond . . .
- Months later, across the wide pond (and the equator)
 - Again, too-short grace periods on a heavily loaded system
 - But more repeatably, within a few hours of stress testing
 - I asked for the .config file, and again got silence
 - Are these guys pranking me???

A Nonsense-Detection Adventure

- Shortly after rcutorture was accepted, I got a strange bug report from across the narrow pond . . .
- Months later, across the wide pond (and the equator)
- A few weeks later...
 - The second bug reporter provides analysis and fix
 - RCU was initializing incorrectly for !HOTPLUG_CPU
 - The synchronize_rcu() function was a fixed-time wait

A Nonsense-Detection Adventure

- Shortly after rcutorture was accepted, I got a strange bug report from across the narrow world
- Months later, across the wide world (and the equator)
- A few weeks later...
 - The second bug report includes analysis and fix
 - RCU was initializing incorrectly for !HOTPLUG_CPU
 - The synchronize_rcu() function was a fixed-time wait

Which exactly explains both bug reports

A Nonsense-Detection Adventure

- Shortly after rcutorture was accepted, I received a strange bug report from a user who was running the narrow version of RCU
- Months later, I received a report from a user (and the equator)
- A few weeks later
 - The second bug report was a duplicate of the first and fix
 - RCU was indeed working correctly CPU
 - The synchronize_rcu() function was a . . . wait

Why did rcutorture miss this instance of nonsense???
Which example explains both?

Why Did rcutorture Miss Nonsense?

- Addition of SUSPEND and HIBERNATE!!!
- These use CPU hotplug to shut down all but one CPU, which then suspends or hibernates
- They force-selected HOTPLUG_CPU
- !HOTPLUG_CPU tests thus built HOTPLUG_CPU
- My tests never tested !HOTPLUG_CPU

Why Did rcutorture Miss Nonsense?

- Addition of SUSPEND and HIBERNATE
- These use CPU hotplug to shut down the CPU, which then suspends the CPU
- They force-select the CPU
- !HOTPLUG_CPU thus built HOTPLUG_CPU
- Never tested !HOTPLUG_CPU

Why Did rcutorture Miss Nonsense?

- Addition of SUSPEND and HIBERNATE
- !HOTPLUG to hotplug to shut CPU, while CPU is running
- They force-set CPU
- !HOTPLUG tests the CPU
- !HOTPLUG never tested !HOTPLUG_C

As the world changes, nonsense changes, and nonsense-finding tests must change

Correctness is not just of the code, but also of the world it runs in!!!

The World Continues to Change!!!

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- Except that they had good reasons to run rcutorture in production!!!

The World Continues to Change!!!

- Someone ran rcutorture in production???
 - And complained about the overhead
- Except that they had good reasons to run rcutorture in production!!!
- So I have patches queued to make this work:
 - nwriters and stall_only module parameters

Summary

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- LLMs are not perfect
- People are not perfect
- The world is not perfect, and always changes
- Our job is nevertheless to make things work!!!

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- LLMs are not perfect
 - People are not perfect
 - The world is not perfect
 - Our job is to make things work!!!
- ways changes

**Which is why we need smart people
such as yourselves!!!**

Questions